# P101 · Unit 8 · Quick reference

Physics, space & scientific thinking
Oscillation and sampled signals
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| | |
|---|---|
| Recall<br>**Cycle and size** | Full cycle: same stage and direction. Amplitude $A$: largest displacement magnitude from equilibrium. Peak-to-peak: $2A$. Offset $y_0$: equilibrium position. |
| Recall<br>**Time and phase** | $f = 1 / T$; $\omega = 2\pi f = 2\pi / T$. $T$ in s, $f$ in Hz, $\omega$ in rad/s. One cycle: $2\pi$ rad. |
| Recall<br>**Sinusoidal model** | $y = y_0 + x$; $x = A\cos(\omega t + \phi)$. Initial phase $\phi$ in rad. Positive maximum at $t_p$: $\phi = -\omega t_p$ modulo $2\pi$. |
| Recall<br>**Supplied SHM rates** | $v = -A\omega\sin(\omega t + \phi)$; $a = -\omega^2 x$. Maximum speed $A\omega$; maximum acceleration magnitude $A\omega^2$. |
| Understand<br>**Spring model** | $F = -kx$, $\omega = \sqrt{k / m}$ , $T = 2\pi\sqrt{m / k}$ . Linear spring, fixed support, negligible damping and spring mass; $k > 0, m > 0$. |
| Understand<br>**Mechanical energy** | $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2$ for the horizontal undamped model.<br>$\lvert v \rvert = \sqrt{(k / m)(A^2 - x^2)}$; direction needs cycle stage. |

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|---|---|
| Understand<br>**Damping and driving** | Free damping removes oscillatory mechanical energy. Steady driving balances loss per cycle. Lightly damped displacement resonance peaks near the undamped natural frequency. |
| Recall<br>**Sample grid** | $\Delta t = 1 / f_s$, $t_n = n / f_s$. Samples per cycle: $f_s / f$. $N$ intervals have $N + 1$ endpoints. Record actual duration. |
| Consult<br>**Upward crossing** | For $x_i \le 0 < x_{i+1}$: $t_c = t_i + \frac{-x_i}{x_{i+1} - x_i}(t_{i+1} - t_i)$. Subtract equilibrium offset first; linear interpolation. |
| Consult<br>**Period estimate** | $\hat{T} = (c_{q-1} - c_0) / (q - 1)$, $q \ge 2$ upward crossings. Inspect individual intervals; too few crossings means no estimate. |
| Understand<br>**Aliasing** | Distinct motions can share samples. Use $f_s > 2f_{\max}$ with a justified signal-frequency bound; equality is unsafe. Graph/crossing estimates need resolution and duration checks. |
| Understand<br>**Evidence** | Model benchmark checks implementation; independent observations test physical predictions. More samples at an unchanged aliasing rate need not resolve ambiguity. |

## Consult · Python

Seconds, metres, radians. Use the checked crossing_period function from Lesson 4.

```
import numpy as np
fs, f, A, phi = 20.0, 2.0, 0.01, 0.0
t = np.arange(81) / fs       # 0 to 4 s inclusive
x = A*np.cos(2*np.pi*f*t + phi)
# Use the lesson's checked crossing_period function:
period, crossings = crossing_period(t, x)
intervals = np.diff(crossings)
error = period - 1/f        # synthetic benchmark only
```

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